

Surgical Wound Infection Rates – Who's the Judge?

A Review of Post-Operative Wound Infection Surveillance Methods

Valerie Humphery R.N., G.D.E. (Melb. Uni)

Background

In 1991, the Australian Council on Healthcare Standards (ACHS), first published a reference manual on 'Clinical Indicators' ⁽¹⁾ for the introduction in 1993 of clinical patient outcome measurements as part of the ACHS Accreditation Programme. Results of Clinical Indicator programmes are compared to provisional 'thresholds' nominated by the ACHS.

An indicator of surgical outcomes is the post-operative wound infection rate, which is defined as 'patients developing an infection of a wound from the fifth post-operative day onwards'.

Wounds are stratified by contamination risk into three classifications of surgery:

- i) Clean
- ii) Contaminated
- iii) Dirty

Wound Infection is defined as 'any surgical wound from which purulent material drains or is obtained. Microbiological confirmation is not necessary'.

The provisional thresholds are:

Clean surgery – 3%

Contaminated surgery – 5%

References for definitions and thresholds are not provided and post discharge surveillance is not a requirement.

This presentation sets out to review the published literature on post-operative wound surveillance methods with a particular view to assessing international comparability of thresholds and definitions proposed by the ACHS.

Introduction:

Two major studies appear to have contributed most to the body of literature in the modern surgical era of post-operative wound surveillance and are cited with almost equal frequency as reference articles.

The first, by Cruse and Foord ⁽²⁾ is a 10 year, prospective study of 62,939 surgical wounds, in a 830 bed, Canadian teaching hospital between 1967 and 1977.

The second, is another 10 year study by Haley et al ⁽³⁾, the 'Study of the Efficacy of Nosocomial Infection Control' (SENIC PROJECT). The one major Australian study on surgical wound infection by McLaws et al ⁽⁴⁾ on 12,742 patients, has more recently joined the other two studies as a much quoted reference source, since publication in 1988.

Methods

A literature search for the period 1981 – 1994, was carried out using the Compact Cambridge CD-ROM. All articles were cross referenced. The variables examined for each study included type of study (incidence or prevalence), number of subjects, risk stratification, wound infection definitions, and infection rates. Only those studies which included over 1,000 cases were reviewed.

Twelve observational studies were reviewed. The Australian study by McLaws et al ⁽⁴⁾ was the only prevalence study. The remainder were incidence studies.

Study Populations and Designs

Comparability of studies is difficult as few authors clearly define inclusion and eligibility criteria for their study populations. Four authors only give accounts of exclusions ^(3, 4, 9, 10). Haley, ⁽³⁾ details patient selection criteria and methods in his SENIC Study. Excluded from the study were federal, state and municipal hospitals, and all Veterans Administration hospitals, thus excluding many medical schools. Cardiac, thoracic, abdominal and caesarean sections were analysed. None of the other specialities were included.

Data collection was by retrospective chart review (RCR), of inpatient records

by some 500 observers. His subsequent 'Nationwide Nosocomial Infection Rate' study ⁽⁵⁾ selected a random sample of 338/4,332 hospitals from SENIC to be included in a target population of 6,449 US hospitals. In contrast, the prospective incidence study by Cruse and Foord ⁽²⁾ included all surgical operations in the hospital, except oral, rectal, vaginal operations; burns and circumcisions. 62,939 'wounds' were included in the study, but it is unclear as to the number of patients.

Other authors have elected to study only 'clean' surgical wounds ^(6,7,8).

Ehrenkranz ⁽⁶⁾ in his report of a multicentre study of seven hospitals and 9,108 cases, defined clean surgery cases: laminectomy, femoropopliteal bypass, inguinal or femoral hernia repair, partial or complete mastectomy, caesarean section, total hip replacement, hip prosthesis and other hip fracture repair. Five hospitals in his study population included all patients with the case definition. However, two hospitals selected only a proportion of the eligible population, potentially threatening the internal validity of the study.

Gil-Egea ⁽⁷⁾ fails to give a case definition for his selection of clean surgery patients and merely lists all the surgical specialities surveyed.

A computer-aided surveillance study by Bremmelgaard et al ⁽⁹⁾ was the only one to specify exclusions of patients for reasons other than surgical procedures: death, re-operation within eight days unless infection discovered before either event.

McLaws et al ⁽²⁾ reported a low 59% participation rate for the Australian study, but gave detailed account for non responders and non participants.

None of the studies gave an account of losses to follow-up.

Wound Infection Definition:

There is a lack on consistency with the case definition for surgical wound

Table 1: Wound Infection Definitions Used in Various Studies

First Author	Year	Definition
Bremmelgaard ⁽⁹⁾	1989	pus at incision, deep infection
Ehrenkranz ⁽⁶⁾	1981	pus, cellulitis, surgeon's diagnosis
Olson ⁽¹³⁾	1990	pus, purulent material on dressing, surgeon's diagnosis
Garibaldi ⁽¹⁸⁾	1991	pus, cellulitis, surgeon's diagnosis
Ferraz ⁽⁸⁾	1992	pus
Cruse ⁽²⁾	1980	pus
Gil-Egea ⁽⁷⁾	1987	pus or exudate, positive micro culture
Simchen ⁽¹⁰⁾	1988	pus, discharge with at least one: systemic antibiotics, local treatment
McLaws ⁽⁴⁾	1988	pus incisional and deep, antibiotics
Haley ⁽¹⁶⁾	1985	pus incisional and deep
Culver ⁽¹⁷⁾	1991	pus incisional and deep, surgeon's diagnosis
Taylor ⁽¹²⁾	1990	pus, incisional and deep, redness drainage and microbiology culture

infection (Table 1). Two studies used the presence of pus alone, with or without positive microbiological culture.

The Centres for Disease Control (CDC) definition for Nosocomial Wound Infections ^(11, 22) includes diagnosis by an attending physician or surgeon and incisional and deep wound infections.

Taylor et al ⁽¹²⁾, studied 3,024 patients in four surgical services, to determine the impact of surgeons' diagnoses on wound infection rates and found that 25% (13/53) clean wound infections were diagnosed by surgeons using clinical impressions alone. There was wide variation in the use of non-standardised criteria. This impact varied across surgical services and between surgeons. In the belief that surgeon diagnosis is a possible confounding variable it is no longer included in their case definition.

However, four studies included clinician diagnosis as definitive of wound infection. ^(6, 15, 17, 18) Ferraz ⁽⁸⁾ in his multihospital Israeli Study, required that the nurse surveyors present their clinical descriptions of the patient to a Panel of four doctors who decided if the wound was infected without seeing the patient.

A more recent British report by Peel and Taylor ⁽²³⁾ recommends that infection be defined as 'a purulent discharge in, or exuding from the wound, or a painful, spreading erythema indicative of cellulitis'.

The SENIC study was the only one to report interobserver reliability.

Contamination Risk Index:

Risk stratification by classification of surgery, has been recommended in the literature since 1964 ⁽¹³⁾. The definitions are included in the CDC National Nosocomial Infections Surveillance System (NNIS) Description of Surveillance Methods ⁽⁹⁾. (NNIS is a US ongoing collaborative study on nosocomial infections sponsored by the CDC).

The four risk categories are: Clean, Clean - contaminated, Contaminated and Dirty and refer to the amount of bacterial contamination potentially present at the time of operation.

Cruse and Foord ⁽²⁾, based their study conclusions on these classifications and until recently, their finding of a 1.5% clean wound infection rate, has been the gold standard for most wound surveillance studies.

Other risk factors reported by Cruse and Foord to reduce wound infection rates were pre-operative shaving with

hexachlorophene, not shaving the operation site, the use of a pre-operative disinfectant to the operative site, and a reduced pre-operative stay. They also reported that the clean wound infection rate doubled with every hour spent in the operating room.

Five studies used all four classical wound definitions ^(2, 9, 16, 17, 18).

McLaws altered the standard definitions and included operations on the lower genital tract and the respiratory tract in clean contaminated surgery. Her definition of contaminated surgery included the dirty classification, as did Olson and colleagues ⁽¹⁵⁾, in their 10 year prospective study of 40,915 patients.

Classification of operations is usually determined at the time of surgery. Few authors have reported who does this. Classifications were determined retrospectively by the investigators of the Australian Study, as the surgical risk definition was not included in the survey questionnaire.

Haley ⁽¹⁶⁾ found that the traditional wound classification system possessed little predictive value after controlling for a simplified risk index. He analysed ten risk factors and developed a model using four indicator variables ⁽¹⁶⁾. The single best predictor was having an operation on the abdomen, followed by surgery of greater than two hours, then by contamination risk index and finally, whether the patient had three or more underlying diagnoses. Variation in his clean wound surgery rates was as low as 1% to as high as 16%. The predictive value was twice as high for the simplified risk index compared to the classical system, 0.67 - 0.36, $P < 0.0001$.

The NNIS have since modified Haley's

risk index and now require risk factor data to be collected on duration of surgery, use of general anaesthesia, the American Society of Anaesthesiologists (ASA) classification (numeric quantification of disease severity), and whether the surgery was an emergency or trauma related.

Culver et al ⁽¹⁷⁾, reported on the results of the NNIS for 1987-1990. Data were reported from 44 hospitals and 84,691 operations. Wound class was still a moderately effective predictor of surgical wound infection.

However as single predictor, ASA classification was equally as good. Infection rates ranged from 1.5% for patients with none of the risk factors to 13% with all risk factors present. There was an increase from 1% to 5.4% amongst patients with clean surgery with an increase in the number of risk factors present.

Other studies have questioned the validity of a single risk factor for wound infection rates. Ferraz ⁽⁸⁾ also cautioned that the classification of clean, carries a potentially inaccurate estimation of patient risk. He found variation in clean wound rates from 4.7% to 21.2% in his study of 1,542 clean operations. He found that the single explanatory variables were age, use of prophylactic antibiotics and surgery type.

Similarly Garibaldi et al ⁽¹⁸⁾, collected prospective data on 1,852 surgical patients and found four variables independent of each other and highly predictive of wound infection. Wound classification, ASA classification, duration of surgery and results of intraoperative microbiology cultures, all had an equally profound impact on surgical wound infection rates.

Infection Rates:

Table 2: Summary of studies reviewed, their risk classifications and clean wound rates (where provided)

First Author	Year	No. Cases	Wound Class	Clean Rate%
Bremmelgaard ⁽⁹⁾	1989	4,340	i-iv	2.3 (0.8-6.1 using risk index)
Ehrenkranz ⁽⁶⁾	1981	9,108	i	1.7-7.9
Olson ⁽¹⁵⁾	1990	40,915	i-iii*	1.8
Garibaldi ⁽¹⁸⁾	1991	1,852	i-iv	3.2 (1-17 using risk index)
Ferraz ⁽⁸⁾	1992	1,542	i	4.7-21.2 (different surgery type specific rates)
Cruse ⁽²⁾	1980	62,939	i-iv	1.5
Gil-Egea ⁽⁷⁾	1987	4,468	i	6.2
Simchen ⁽¹⁰⁾	1988	5,571	unknown	
McLaws ⁽⁴⁾	1988	2,742	i-iii*	4.8 (95% CI 4.3-5.4)
Haley ⁽¹⁶⁾	1985	59,352	i-iv	2.9 (1.1-15.8 using risk index)
Culver ⁽¹⁷⁾	1991	84,691	i-iv	2.1 (1-5.4 using risk index)
Taylor ⁽¹²⁾	1990	3,024	?	?

* Denotes risk classification definitions which have been altered for the study in question

No references to the ACHS 3% threshold for clean surgery, could be found in any of the articles reviewed or cross-referenced. Rates are not comparable between countries, hospitals or surgeons as many baselines exist. Table 2

Follow-up time for surveillance varied from inpatient surveillance only to 30 days post-operatively. Post-discharge surveillance was either not included or non-standardised. Five studies ^(2, 8, 15, 9, 10) included post-discharge rates in the numerators.

Olson and Ferraz ^(8, 15) followed patients through hospital Outpatients Services; Cruse and Foord ⁽²⁾ by telephone and Bremmelgaard ⁽⁹⁾ and Simchen ⁽¹⁰⁾ by mail. Response rates were not validated.

In recent review by Holz and Wenzel ⁽¹⁹⁾, of post-discharge surveillance, it was found that 20% - 70% of post-surgical wound infections do not become apparent until after discharge.

Reimer and Nicholle, ⁽²⁰⁾ found that most post-discharge infections were 'clean' surgery and early discharges.

ANALYSIS OF DATA:

The Israeli Study by Simchen ⁽¹⁰⁾ did not attempt to produce wound infection rates, but rather set out to standardise the surveillance system by validating the reliability of methods and definitions. The purpose of the validation studies was to demonstrate uniformity and accuracy, that would justify interhospital comparisons. Haley reviewed the study design and criticised it somewhat harshly. ⁽²¹⁾

Cruse and Foord, ⁽²⁾ whilst still being quoted as the reference for many

studies, failed to give any statistical interpretation of their results.

Ehrenkranz ⁽⁶⁾ stratified patients by remote infection, operation lasting more than two hours, and diabetes mellitus, to determine high and low risk groups of prospective operations.

Bremmelgaard ⁽⁹⁾ reclassified his population into three atomic groups when he found that his statistical model did not work with the surgical departments under review. Infection rates in his clean surgery ranged from 0.8% to 6.1%, depending on the presence of five significant risk factors: wound classification, length of operation, age, department, and date of operation.

Garibaldi et al ⁽¹⁸⁾, found that there were four variables independent of each other: wound classification, length of operation, ASA class, results of intraoperative cultures. The unadjusted predicted probability for infection of clean surgery was 2.2% and for clean surgery with high ASA codes and surgery of long duration, the predicted probability rose to 8.3%.

Haley et al ⁽¹⁶⁾ developed a single multivariate index of patient susceptibility using the SENIC study population from 1970 and 1975 - 1976

McLaws ⁽²⁾ and colleagues employed a multiple logistic regression model to measure outcomes between each predictor and infection independently.

Almost all investigators who have used multivariate analyses to select the most significant risk factors for surgical wound infection, have included a marker of host susceptibility, wound contamination and duration of surgery.

SUMMARY:

Surgical wound surveillance is deemed to be necessary, if wound infections are to be reduced. Few studies have given a plausible account of patient selection, losses to follow-up or validation of methods. Comparison of studies is virtually impossible when so many methodological inconsistencies exist.

Surgeon defined infections are included in the NNIS definitions, yet it has been shown that surgeon-defined rates can markedly skew infection rates.

The SENIC project whilst laudible in its size and detail omits to clarify some important methodological issues. Many hospitals, operations and patients were excluded from the database.

Reports that give infection rates without stratification for surgical procedure and severity of illness are uninterpretable, as shown by the wide variation in results in studies on clean wound rates.

Even with stratification, there is variation within the four classical definitions as demonstrated by McLaws et al. ⁽⁴⁾

The inclusion of post-discharge rates in some studies has shown to significantly alter infection rates and is a source of possible confounding.

The most sophisticated data analysis cannot validate a study that is poorly designed and executed in the first instance.

Evidence would suggest that new designs and methods are needed for the 1990s, that adequately reflect the shorter length of post-operative stays and complexity of intrinsic risk factors.

CONCLUSION:

It is debatable whether interhospital comparisons of surgical wound infection rates in Australia can be meaningful and, even more unsettling, that the threshold rates can be set by ACHS with little validation. Casemix peculiarities will mean nonuniformity for comparison of overall rates as will anomalies such as different sizes of hospital populations, retrospective data versus prospective data and incidence studies versus prevalence studies.

The definition of a surgical wound infection should be broadened to be consistent with the CDC definitions of surgical site infections, which give cognisance to both incisional and deep wound infections.⁽²²⁾

The modified ACHS definitions for wound class, disallows even a modest comparison with international studies.

It could be argued that, as there are no published reports on the accuracy or reproducibility of the original definitions, this is not important, but creating a new set of definitions nevertheless requires validation.

Economic constraints have necessitated shorter hospital stays and the introduction of 5 day surgical services. Post-operative observation of wounds is constrained by the early discharge of patients. The majority of infections will not be apparent in this time frame and a cost-effective standardised method of post-discharge surveillance would need to be implemented.

The ACHS inclusion criteria for patients More than 5 days post surgery, may mean the deletion of approximately 50% of cases from the denominator. No other study has reported this exclusion.

A composite risk index including wound classifications, an indicator of patient susceptibility and duration of surgery, should be required before meaningful comparisons can be made between hospitals, surgeons and procedures. Results would then perhaps be more convincing as an outcome of quality care.

Target populations, included and excluded operations should be clearly defined and patients stratified within each surgical procedure.

It would be hoped that surveillance programmes may be beneficial to the patient and surgeon by providing an objective, specific internal audit tool.

This literature review has found no comparable definitions or study designs to support the current recommended

ACHS definitions and study design.

It is a retrograde step to use definitions and thresholds defined in the 1980's for surveillance activities in the 1990's – who's the judge of our wound infection rates?

References:

- (1) Clinical Indicators: Users' Manual: Hospital-Wide Medical Indicators 1991. Australian Council of Healthcare Standards (ACHS), July 1991.
- (2) Cruse PJE, Foord R. The epidemiology of wound infections. A ten-year prospective study of 62,939 wounds. Surg Clin North Am 1980; 60: 27-40.
- (3) Haley RW, Quade D, Freeman HE; et al. Study of the efficacy of nosocomial infection control (SENIC project): summary of study design. Am J Epidemiol 1980; 111: 472-485.
- (4) McLaws M, Irwig LM, Mock P, Berry G, Gold J. Predictors of surgical wound infection in Australia: a national study. MJ Aust 1988; 149: 591-595.
- (5) Haley RW, Culver DH, White JW, Morgan WM, Emori TJ. The national nosocomial infection rate: a new need for vital statistics. Am J Epidemiol 1985; 121: 159-167.
- (6) Ehrenkranz NJ. Surgical wound infections in clean operations: risk stratification for interhospital comparisons. Am J Med 1981; 70: 909-914.
- (7) Gil-Egea MJ, Pi-Sunyer MT, Verdaguer; et al. Surgical wound infections prospective study of 4,468 clean wounds. Infect Control 1987; 8: 277-280.
- (8) Ferraz EM, Bacelar TS, Aguiar AAB; et al. Wound infection rates in clean surgery: a potentially misleading risk classification. Infect Control Hosp Epidemiol 1992; 13: 457-462.
- (9) Bremmelgaard A, Raalave D, Beier-Holgersen R; et al. Computer-aided surveillance of surgical infections and identification of risk factors. J of Hosp Infect 1989; 13: 457-462.
- (10) Simchen E, Wax Y, Pevsner B, Erdal M; et al. The Israeli study of surgical infections (ISSI): Methods for developing a standardized surveillance system for a multi-centre study of surgical infections. Infect Control Hosp Epidemiol 1988; 9: 232-240.
- (11) Garner JS, Jarvis WR, Emori TJ, Horan TC, Hughes JM. CDC definitions for nosocomial infections. Am J Infect Control 1988; 16: 128-140.
- (12) Taylor G, McKenzie M, Kirkland T, Wiens R. Effect of surgeon's diagnosis on surgical wound infection rates. Am J Infect Control 1990; 18: 295-299.
- (13) National Academy of Sciences - National Research Council, Division of Medical Sciences, Ad Hoc Committee of the Committee of Trauma: Post-operative wound infections. The influence of ultraviolet irradiation on the operating room and various other factors. Ann Surg 1964; 160 (Suppl2): 1.
- (14) Emori TJ, Culver DH, Horan TC, Jarvis WR; et al. National nosocomial infections surveillance systems (NNIS): description of surveillance methods. Am J Infect Control 1991; 19: 19-35.
- (15) Olson M, Lee JT. Continuous, 10-year wound infection surveillance: results, advantages and unanswered questions. Arch Surg 1990; 125: 794-803.
- (16) Haley RW, Culver DH, Morgan WM, White JW, Emori TJ, Hooton TM. Identifying patients at risk of surgical wound infection: a simple multivariate index of patient susceptibility and wound contamination. Am J Epidemiol 1985; 121: 206-215.
- (17) Culver DH, Horan TC, Gaynes RP; et al. Surgical wound infection rates by wound class, operative procedure and patient risk index. Am J Med 1991; 91 (Suppl 3B): 3b-152s-3b-157s.
- (18) Garibaldi RA, Cushin D, Lerer T. Risk Factors for postoperative infection. Am J Med 1991; 91: (Suppl 3B): 3b-158s-3b-163s.
- (19) Holtz TH, Wenzel R. Postdischarge surveillance for nosocomial wound infection: A brief review and commentary. Am J Infect Control 1992; 20: 206-213.
- (20) Reimer K, Nicolle LE. The impact of post discharge infection on surgical wound infection rates. Infect Control 1987; 8: 237-240.
- (21) Haley RW. The Vicissitudes of prospective multihospital surveillance studies: the Israeli study of surgical infections. Infect Control Hosp Epidemiol 1988; 9: 228-231.
- (22) Horan TC, Gaynes RP, Martone WJ, Jarvis WR, Emori TG. CDC definitions of nosocomial surgical site infections, 1992: A modification of CDC definitions of surgical wound infections. Infect Control Hosp Epidemiol 1992; 20: 271-274.
- (23) Peel ALG, Taylor AW. Proposed definitions for the audit of postoperative infection: a discussion paper. Ann Roy Coll Surg Engl 1991; 73: 385-388.